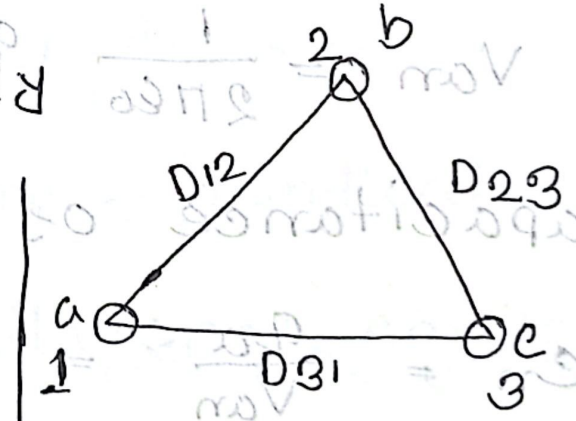
Lecture - 06
21.07.26

Art - 3.5 : Capacitance of a 3-ø line with unsymmetrical spacing

Assume, that line is fully transposed

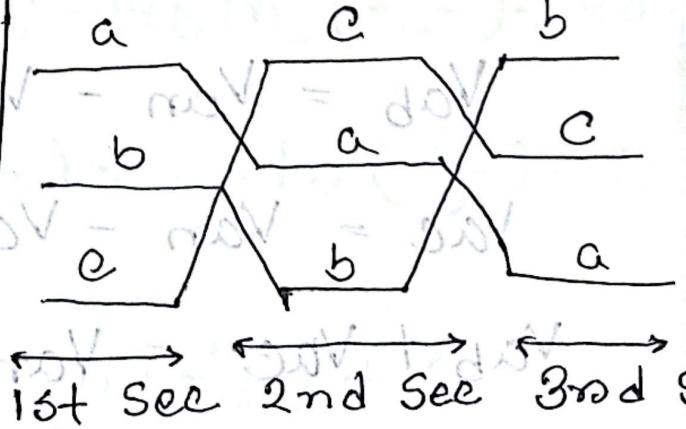


1st section : $V_{ab1} =$

$$\frac{1}{2\pi k}\left[q_{a1}\ln\left(\frac{D12}{r}\right) + q_{b1}\ln\left(\frac{r}{D12}\right) + q_{c1}\ln\left(\frac{D23}{D31}\right)\right]$$

2nd section : $V_{ab2} =$

$$\frac{1}{2\pi k}\left[q_{a2}\ln\left(\frac{D23}{r}\right) + q_{b2}\ln\left(\frac{r}{D23}\right) + \right.$$



Jannat

sat sun mon tue wed thu fri Date :

$q_{c2} \ln\left(\frac{D_{31}}{D_{12}}\right)$]

3rd section : $V_{ab3} = \frac{1}{2\pi k}\left[q_{a3} \ln\left(\frac{D_{31}}{r}\right) + q_{b3} \ln\left(\frac{r}{D_{31}}\right) + q_{c3} \ln\left(\frac{D_{12}}{D_{23}}\right)\right]$

Assuming,

$q_{a1} = q_{a2} = q_{a3} = q_a$

$q_{b1} = q_{b2} = q_{b3} = q_b$

$q_{c1} = q_{c2} = q_{c3} = q_c$

$$V_{ab(avg)} = \frac{V_{ab1} + V_{ab2} + V_{ab3}}{3}$$

$$= \frac{1}{6\pi k}\left[q_a \ln\left(\frac{D_{12} D_{23} D_{31}}{r^3}\right) + q_b \ln\left(\frac{r^3}{D_{12} D_{23} D_{31}}\right) + q_3 \ln\left(\frac{D_{12} D_{23} D_{31}}{D_{12} D_{23} D_{31}}\right)\right.$$

$$V_{ab(avg)} = \frac{1}{2\pi k}\left[q_a \ln\left(\frac{D_{eq}}{r}\right) + q_b \ln\left(\frac{r}{D_{eq}}\right)\right]$$

Here, $D_{eq} = (D_{12} D_{23} D_{31})^{1/3}$

Similarly,

$$V_{ac} = \frac{1}{2\pi k}\left[q_a \ln\left(\frac{D_{eq}}{r}\right) + q_c \ln\left(\frac{r}{D_{eq}}\right)\right]$$

$$V_{ab} + V_{ac} = \frac{1}{2\pi k}\left[2q_a \ln\left(\frac{D_{eq}}{r}\right) + (q_b + q_c) \ln\left(\frac{r}{D_{eq}}\right)\right]$$

As, $q_a + q_b + q_c = 0$.

$$q_b + q_c = -q_a$$

sat sun mon tue wed thu fri Date :

$$V_{ab} + V_{ac} = 3V_{an}$$

$$\text{So, } V_{an} = \frac{q_a}{2\pi k} \ln\left(\frac{D_{eq}}{r}\right)$$

The capacitance of line to neutral of the transposed line

$$C_n = \frac{q_a}{V_{an}} = \frac{2\pi k}{\ln\left(\frac{D_{eq}}{r}\right)} \text{ F/m to neutral}$$

For Air medium, $k_r = 1$

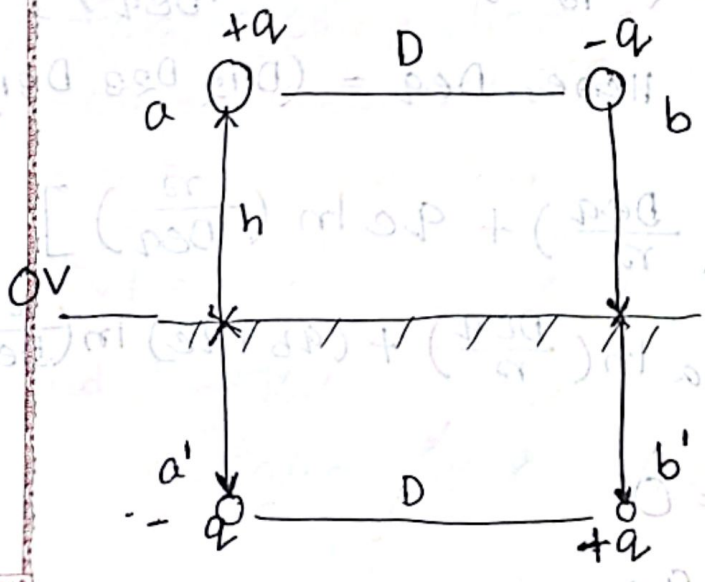
$$C_n = \frac{0.0242}{\log\left(\frac{D_{eq}}{r}\right)} \mu F/km \text{ to neutral}$$

$$I_a \text{ (line charging)} = j\omega C_n V_{an} \quad A/km$$

3.7.1

Method of Image

1 ø line



opposite ও same distance ও image imagine করা হয়

Image Conductor

sat sun mon tue wed thu fri Date:

$$V_{ab} = \frac{1}{2\pi k}\left[q_a \ln\left(\frac{D}{r}\right) + q_b \ln\left(\frac{r}{D}\right) + q_a' \ln\left(\frac{\sqrt{4h^2+D^2}}{2h}\right) + q_b' \ln\left(\frac{2h}{\sqrt{4h^2+D^2}}\right)\right]$$

$$= \frac{q}{\pi k} \ln\left(\frac{2hD}{r\sqrt{4h^2+D^2}}\right)$$

$$C_{ab} = \frac{\pi k}{\ln \frac{D}{r(1+(D^2/4h^2))^{1/2}}} \text{ F/m line to line}$$

$$C_n = \frac{2\pi k}{\ln \frac{D}{r(1+(D^2+4h^2))^{1/2}}}$$

* Presence of earth modifies the radius r to $r(1+(D^2+4h^2))^{1/2}$

For 'h' large compared to D the effect of earth on line capacitance is of negligible order.

Example - 3.3 → Not necessary (Mohona Mam)

3.1, 3.2 → Must

- Avoid extremely complex
- Ant 3.1 to 3.7